

PROJECT ASSIGNMENT 2

Group10-PA2 User Analysis

Clearly labeled simulated brainstorming, affinity analysis, voting, and prioritization

FIFA.com and Chess.com

Course	CSC13112 - UI/UX Design, FIT-HCMUS
Group	Group10
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Scope	Clearly labeled simulated brainstorming, affinity analysis, voting, and prioritization

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1. Analysis boundary and source discipline

Simulated brainstorming record: The workshop, raw-note generation, votes, discussion, attendance, and consensus below are scenario-based synthetic evidence prepared for course reporting. They are not contemporaneous meeting evidence.

Screen evidence: Visible interface states seed observable notes only.

Simulated study: Participant-behavior notes and votes are synthetic scenario records.

Design inference: Clusters and priorities are interpretations that require PA3 testing.

2. Simulated brainstorming session

Field	Record
Record type	Simulated brainstorming record
Date and duration	22 Jul 2026, 19:30-20:45 (75 minutes)
Participants	Le Minh; Nguyen Vu Bach; Pham Nguyen Gia Bao; Trang Minh Nhut
Objective	Transform PA1 continuity, screen evidence, and simulated-study patterns into prioritized PA2 problems.
Silent note generation	12 minutes; each member drafts atomic notes independently.
Round-robin sharing	16 minutes; one note per turn, no evaluation during sharing.
Clustering	18 minutes; group by user task and breakdown rather than feature.
Duplicate merging	8 minutes; retain the clearest atomic wording and all source tags.
Cluster naming	6 minutes; name clusters with user-facing task language.
Prioritization	8 minutes; six simulated votes per member per product.
Consensus	5 minutes; review weighted scores and resolve the selected problem statement.
Owner assignment	2 minutes; Bach owns FIFA synthesis, Bao owns Chess synthesis, Nhut owns diagrams/HCI mapping, Le owns integration/traceability.

3. Raw notes: 36 FIFA and 36 Chess

Record 1

Note ID	F-RN-01
Product	FIFA
Cluster	Ticket status clarity
Atomic note	Current sale state needs one label
Source type	screen evidence

Record 2

Note ID	F-RN-02
Product	FIFA
Cluster	Ticket status clarity
Atomic note	Missing state differs from sold out
Source type	simulated participant behavior

Record 3

Note ID	F-RN-03
Product	FIFA
Cluster	Ticket status clarity
Atomic note	Resale must not look like primary sale
Source type	PA1 inherited finding

Record 4

Note ID	F-RN-04
Product	FIFA
Cluster	Ticket status clarity
Atomic note	Waiting room is a temporary state
Source type	design inference

Record 5

Note ID	F-RN-05
Product	FIFA
Cluster	Ticket status clarity
Atomic note	Hospitality is a distinct option
Source type	screen evidence

Record 6

Note ID	F-RN-06
Product	FIFA
Cluster	Ticket status clarity
Atomic note	State comparison belongs in one view
Source type	simulated participant behavior

Record 7

Note ID	F-RN-07
Product	FIFA
Cluster	Handoff trust
Atomic note	Preview the destination domain
Source type	PA1 inherited finding

Record 8

Note ID	F-RN-08
Product	FIFA
Cluster	Handoff trust
Atomic note	Explain why departure is needed
Source type	design inference

Record 9

Note ID	F-RN-09
Product	FIFA
Cluster	Handoff trust
Atomic note	Offer stay on FIFA.com
Source type	screen evidence

Record 10

Note ID	F-RN-10
Product	FIFA
Cluster	Handoff trust
Atomic note	Preserve a safe return route
Source type	simulated participant behavior

Record 11

Note ID	F-RN-11
Product	FIFA
Cluster	Handoff trust
Atomic note	Do not imply partner transaction status
Source type	PA1 inherited finding

Record 12

Note ID	F-RN-12
Product	FIFA
Cluster	Handoff trust
Atomic note	Keep tournament context through handoff
Source type	design inference

Record 13

Note ID	F-RN-13
Product	FIFA
Cluster	Mobile overview
Atomic note	Keep selected tournament first
Source type	screen evidence

Record 14

Note ID	F-RN-14
Product	FIFA
Cluster	Mobile overview
Atomic note	Avoid a long undifferentiated card stack
Source type	simulated participant behavior

Record 15

Note ID	F-RN-15
Product	FIFA
Cluster	Mobile overview
Atomic note	Restore filters on return
Source type	PA1 inherited finding

Record 16

Note ID	F-RN-16
Product	FIFA
Cluster	Mobile overview
Atomic note	Use labels that survive small screens
Source type	design inference

Record 17

Note ID	F-RN-17
Product	FIFA
Cluster	Mobile overview
Atomic note	Make status scannable without color
Source type	screen evidence

Record 18

Note ID	F-RN-18
Product	FIFA
Cluster	Mobile overview
Atomic note	Resume with one recognition step
Source type	simulated participant behavior

Record 19

Note ID	F-RN-19
Product	FIFA
Cluster	Browse load
Atomic note	Ticket entry competes with content
Source type	PA1 inherited finding

Record 20

Note ID	F-RN-20
Product	FIFA
Cluster	Browse load
Atomic note	Repeated cards increase scanning
Source type	design inference

Record 21

Note ID	F-RN-21
Product	FIFA
Cluster	Browse load
Atomic note	Tournament identity should anchor actions
Source type	screen evidence

Record 22

Note ID	F-RN-22
Product	FIFA
Cluster	Browse load
Atomic note	Reduce cross-page comparison
Source type	simulated participant behavior

Record 23

Note ID	F-RN-23
Product	FIFA
Cluster	Browse load
Atomic note	Keep action wording consistent
Source type	PA1 inherited finding

Record 24

Note ID	F-RN-24
Product	FIFA
Cluster	Browse load
Atomic note	Separate browse from purchase intent
Source type	design inference

Record 25

Note ID	F-RN-25
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Show last-updated time
Source type	screen evidence

Record 26

Note ID	F-RN-26
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Name the governed source
Source type	simulated participant behavior

Record 27

Note ID	F-RN-27
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Label stale data explicitly
Source type	PA1 inherited finding

Record 28

Note ID	F-RN-28
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Allow relevant change alerts
Source type	design inference

Record 29

Note ID	F-RN-29
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Explain notification permission
Source type	screen evidence

Record 30

Note ID	F-RN-30
Product	FIFA
Cluster	Freshness and alerts
Atomic note	Let users edit or stop alerts
Source type	simulated participant behavior

Record 31

Note ID	F-RN-31
Product	FIFA
Cluster	Recovery and return
Atomic note	Retry missing state without losing context
Source type	PA1 inherited finding

Record 32

Note ID	F-RN-32
Product	FIFA
Cluster	Recovery and return
Atomic note	Network loss should retain the last snapshot
Source type	design inference

Record 33

Note ID	F-RN-33
Product	FIFA
Cluster	Recovery and return
Atomic note	Login mismatch needs safe recovery
Source type	screen evidence

Record 34

Note ID	F-RN-34
Product	FIFA
Cluster	Recovery and return
Atomic note	Expired return links need explanation
Source type	simulated participant behavior

Record 35

Note ID	F-RN-35
Product	FIFA
Cluster	Recovery and return
Atomic note	Partner failure must offer stay
Source type	PA1 inherited finding

Record 36

Note ID	F-RN-36
Product	FIFA
Cluster	Recovery and return
Atomic note	Never infer purchase completion
Source type	design inference

Record 37

Note ID	C-RN-01
Product	Chess
Cluster	Entry choice overload
Atomic note	Beginner review needs one named entry
Source type	screen evidence

Record 38

Note ID	C-RN-02
Product	Chess
Cluster	Entry choice overload
Atomic note	Analysis setup paths compete
Source type	simulated participant behavior

Record 39

Note ID	C-RN-03
Product	Chess
Cluster	Entry choice overload
Atomic note	Feature names require recall
Source type	PA1 inherited finding

Record 40

Note ID	C-RN-04
Product	Chess
Cluster	Entry choice overload
Atomic note	Completed game should expose review
Source type	design inference

Record 41

Note ID	C-RN-05
Product	Chess
Cluster	Entry choice overload
Atomic note	Expert view opens too early
Source type	screen evidence

Record 42

Note ID	C-RN-06
Product	Chess
Cluster	Entry choice overload
Atomic note	Outcome labels are easier to recognize
Source type	simulated participant behavior

Record 43

Note ID	C-RN-07
Product	Chess
Cluster	Learning priority
Atomic note	Show one important mistake first
Source type	PA1 inherited finding

Record 44

Note ID	C-RN-08
Product	Chess
Cluster	Learning priority
Atomic note	Explain why before numeric depth
Source type	design inference

Record 45

Note ID	C-RN-09
Product	Chess
Cluster	Learning priority
Atomic note	Make the next action explicit
Source type	screen evidence

Record 46

Note ID	C-RN-10
Product	Chess
Cluster	Learning priority
Atomic note	Keep the lesson goal visible
Source type	simulated participant behavior

Record 47

Note ID	C-RN-11
Product	Chess
Cluster	Learning priority
Atomic note	Do not present every candidate equally
Source type	PA1 inherited finding

Record 48

Note ID	C-RN-12
Product	Chess
Cluster	Review setup
Atomic note	No game needs a clear setup route
Source type	design inference

Record 49

Note ID	C-RN-13
Product	Chess
Cluster	Review setup
Atomic note	Invalid PGN must preserve input
Source type	screen evidence

Record 50

Note ID	C-RN-14
Product	Chess
Cluster	Review setup
Atomic note	Board orientation needs confirmation
Source type	simulated participant behavior

Record 51

Note ID	C-RN-15
Product	Chess
Cluster	Review setup
Atomic note	Engine failure needs a safe fallback
Source type	PA1 inherited finding

Record 52

Note ID	C-RN-16
Product	Chess
Cluster	Review setup
Atomic note	Premium restriction needs honest scope
Source type	design inference

Record 53

Note ID	C-RN-17
Product	Chess
Cluster	Vocabulary load
Atomic note	Evaluation terms need inline help
Source type	screen evidence

Record 54

Note ID	C-RN-18
Product	Chess
Cluster	Vocabulary load
Atomic note	Plain language should lead
Source type	simulated participant behavior

Record 55

Note ID	C-RN-19
Product	Chess
Cluster	Vocabulary load
Atomic note	Terminology help should not interrupt
Source type	PA1 inherited finding

Record 56

Note ID	C-RN-20
Product	Chess
Cluster	Vocabulary load
Atomic note	Returning learners forget feature names
Source type	design inference

Record 57

Note ID	C-RN-21
Product	Chess
Cluster	Vocabulary load
Atomic note	Recognition should replace recall
Source type	screen evidence

Record 58

Note ID	C-RN-22
Product	Chess
Cluster	Mobile scanning
Atomic note	Keep the board and main mistake together
Source type	simulated participant behavior

Record 59

Note ID	C-RN-23
Product	Chess
Cluster	Mobile scanning
Atomic note	Collapse secondary panels
Source type	PA1 inherited finding

Record 60

Note ID	C-RN-24
Product	Chess
Cluster	Mobile scanning
Atomic note	Use color-independent indicators
Source type	design inference

Record 61

Note ID	C-RN-25
Product	Chess
Cluster	Mobile scanning
Atomic note	Preserve state after interruption
Source type	screen evidence

Record 62

Note ID	C-RN-26
Product	Chess
Cluster	Mobile scanning
Atomic note	Maintain readable tap targets
Source type	simulated participant behavior

Record 63

Note ID	C-RN-27
Product	Chess
Cluster	Practice continuation
Atomic note	Link the mistake to one lesson
Source type	PA1 inherited finding

Record 64

Note ID	C-RN-28
Product	Chess
Cluster	Practice continuation
Atomic note	Offer one relevant puzzle
Source type	design inference

Record 65

Note ID	C-RN-29
Product	Chess
Cluster	Practice continuation
Atomic note	Preserve the reviewed position
Source type	screen evidence

Record 66

Note ID	C-RN-30
Product	Chess
Cluster	Practice continuation
Atomic note	Let users continue later
Source type	simulated participant behavior

Record 67

Note ID	C-RN-31
Product	Chess
Cluster	Practice continuation
Atomic note	Explain why the practice is relevant
Source type	PA1 inherited finding

Record 68

Note ID	C-RN-32
Product	Chess
Cluster	Advanced depth disclosure
Atomic note	Reveal engine detail on demand
Source type	design inference

Record 69

Note ID	C-RN-33
Product	Chess
Cluster	Advanced depth disclosure
Atomic note	Do not remove expert access
Source type	screen evidence

Record 70

Note ID	C-RN-34
Product	Chess
Cluster	Advanced depth disclosure
Atomic note	Remember the disclosure choice
Source type	simulated participant behavior

Record 71

Note ID	C-RN-35
Product	Chess
Cluster	Advanced depth disclosure
Atomic note	Keep beginner explanation stable
Source type	PA1 inherited finding

Record 72

Note ID	C-RN-36
Product	Chess
Cluster	Advanced depth disclosure
Atomic note	Separate learning priority from depth
Source type	design inference

Scenario-based synthetic evidence: All 72 notes are one-idea records; simulated participant behavior notes are not real observation claims.

4. Affinity diagrams and cluster synthesis

Record 1

Product	FIFA
Cluster	Ticket status clarity
Count	6
Representative notes	Current sale state needs one label; Missing state differs from sold out
Pattern	Repeated task-state or orientation breakdown
Affected task	Compare ticket states
Severity	Critical
Design direction	One comparison dashboard

Record 2

Product	FIFA
Cluster	Handoff trust
Count	6
Representative notes	Preview the destination domain; Explain why departure is needed
Pattern	Repeated task-state or orientation breakdown
Affected task	Preview destination
Severity	High
Design direction	Destination preview + stay/return

Record 3

Product	FIFA
Cluster	Mobile overview
Count	6
Representative notes	Keep selected tournament first; Avoid a long undifferentiated card stack
Pattern	Repeated task-state or orientation breakdown
Affected task	Resume on mobile
Severity	Medium
Design direction	Persist tournament context

Record 4

Product	FIFA
Cluster	Browse load
Count	6
Representative notes	Ticket entry competes with content; Repeated cards increase scanning
Pattern	Repeated task-state or orientation breakdown
Affected task	Find ticket entry
Severity	Medium
Design direction	Task-led entry

Record 5

Product	FIFA
Cluster	Freshness and alerts
Count	6
Representative notes	Show last-updated time; Name the governed source
Pattern	Repeated task-state or orientation breakdown
Affected task	Inspect freshness / subscribe
Severity	Medium
Design direction	Timestamp + controlled alerts

Record 6

Product	FIFA
Cluster	Recovery and return
Count	6
Representative notes	Retry missing state without losing context; Network loss should retain the last snapshot
Pattern	Repeated task-state or orientation breakdown
Affected task	Recover and return
Severity	Medium
Design direction	State-specific recovery

Record 7

Product	Chess
Cluster	Entry choice overload
Count	6
Representative notes	Beginner review needs one named entry; Analysis setup paths compete
Pattern	Repeated task-state or orientation breakdown
Affected task	Open review
Severity	Critical
Design direction	Beginner review preset

Record 8

Product	Chess
Cluster	Learning priority
Count	5
Representative notes	Show one important mistake first; Explain why before numeric depth
Pattern	Repeated task-state or orientation breakdown
Affected task	Identify mistake
Severity	High
Design direction	One mistake first

Record 9

Product	Chess
Cluster	Review setup
Count	5
Representative notes	No game needs a clear setup route; Invalid PGN must preserve input
Pattern	Repeated task-state or orientation breakdown
Affected task	Open game
Severity	Medium
Design direction	Validated setup states

Record 10

Product	Chess
Cluster	Vocabulary load
Count	5
Representative notes	Evaluation terms need inline help; Plain language should lead
Pattern	Repeated task-state or orientation breakdown
Affected task	Read explanation
Severity	High
Design direction	Inline plain-language help

Record 11

Product	Chess
Cluster	Mobile scanning
Count	5
Representative notes	Keep the board and main mistake together; Collapse secondary panels
Pattern	Repeated task-state or orientation breakdown
Affected task	Review on mobile
Severity	Medium
Design direction	Primary-card hierarchy

Record 12

Product	Chess
Cluster	Practice continuation
Count	5
Representative notes	Link the mistake to one lesson; Offer one relevant puzzle
Pattern	Repeated task-state or orientation breakdown
Affected task	Continue practice
Severity	High
Design direction	One relevant next exercise

Record 13

Product	Chess
Cluster	Advanced depth disclosure
Count	5
Representative notes	Reveal engine detail on demand; Do not remove expert access
Pattern	Repeated task-state or orientation breakdown
Affected task	Reveal detail
Severity	Medium
Design direction	Progressive disclosure

5. Simulated dot voting

Simulated brainstorming record: Each member receives six votes for FIFA and six for Chess; totals equal 24 votes per product.

Record 1

Member	Le Minh
Status	3
Handoff	2
Mobile	0
Browse	0
Freshness	1
Recovery	0
Total	6

Record 2

Member	Nguyen Vu Bach
Status	3
Handoff	2
Mobile	0
Browse	0
Freshness	0
Recovery	1
Total	6

Record 3

Member	Pham Nguyen Gia Bao
Status	2
Handoff	2
Mobile	0
Browse	1
Freshness	1
Recovery	0
Total	6

Record 4

Member	Trang Minh Nhut
Status	2
Handoff	2
Mobile	1
Browse	0
Freshness	1
Recovery	0
Total	6

FIFA cluster	Total votes
Ticket status clarity	10
Handoff trust	8
Freshness and alerts	3
Mobile overview	1
Browse load	1
Recovery and return	1

Record 1

Member	Le Minh
Entry	1
Priority	0
Setup	0
Vocabulary	2
Mobile	0
Practice	2
Depth	1
Total	6

Record 2

Member	Nguyen Vu Bach
Entry	1
Priority	1
Setup	1
Vocabulary	1
Mobile	0
Practice	1
Depth	1
Total	6

Record 3

Member	Pham Nguyen Gia Bao
Entry	1
Priority	1
Setup	1
Vocabulary	1
Mobile	0
Practice	2
Depth	0
Total	6

Record 4

Member	Trang Minh Nhut
Entry	1
Priority	1
Setup	1
Vocabulary	1
Mobile	1
Practice	0
Depth	1
Total	6

Chess cluster	Total votes
Vocabulary load	5
Practice continuation	5
Entry choice overload	4
Learning priority	3
Review setup	3
Advanced depth disclosure	3
Mobile scanning	1

Field	Record
Discussion notes	FIFA votes concentrate on status and handoff confidence. Chess votes split across vocabulary and continuation, while entry overload is the enabling problem.
Tie-break rule	If totals tie, select the item with higher severity, then stronger traceability to the PA3 prototype task.

Field	Record
Consensus statement	Prioritize FIFA ticket status plus handoff confidence, and treat Chess entry, explanation, and continuation as one guided beginner review problem.
Selected problems	FIFA: ticket status clarity and handoff trust. Chess: guided beginner review.

6. Weighted prioritization

Formula: Weighted score = Severity x 25% + Task frequency x 15% + Risk x 20% + Reach x 10% + Evidence strength x 15% + PA3 relevance x 15%. Each factor is scored 1-5; maximum is 5.00.

Record 1

Candidate	FIFA status + freshness
Severity	5
Frequency	4
Risk	5
Reach	4
Evidence	4
PA3	5
Weighted	4.60
Score rationale	Critical ambiguity; frequent planning; trust risk; broad reach; strong screen/synthetic trace; direct PA3 module.

Record 2

Candidate	FIFA handoff confidence
Severity	5
Frequency	3
Risk	5
Reach	4
Evidence	4
PA3	5
Weighted	4.45
Score rationale	Critical departure risk; lower frequency; broad consequence; direct PA3 decision.

Record 3

Candidate	FIFA mobile overview
Severity	3
Frequency	4
Risk	3
Reach	4
Evidence	4
PA3	3
Weighted	3.50
Score rationale	Frequent but lower consequence; evidence supports orientation rather than transaction risk.

Record 4

Candidate	Chess guided beginner review
Severity	5
Frequency	5
Risk	4
Reach	5
Evidence	4
PA3	5
Weighted	4.65
Score rationale	High learning severity/frequency/reach; strong PA3 relevance; synthetic outcomes require validation.

Record 5

Candidate	Chess mobile scanning
Severity	3
Frequency	4
Risk	3
Reach	4
Evidence	3
PA3	3
Weighted	3.35
Score rationale	Broad but secondary to entry and learning-priority breakdown.

Record 6

Candidate	Chess advanced depth
Severity	3
Frequency	2
Risk	3
Reach	3
Evidence	3
PA3	4
Weighted	3.00
Score rationale	Important control but lower beginner frequency; retained as a disclosure requirement.

7. Tough problems**7.1 TP-FIFA**

TP-FIFA: Users need to compare current ticket state, freshness and official destination before leaving FIFA.com.

Field	Record
Affected users	Trust-sensitive planners and tournament followers
Context	Ticket planning before an outbound handoff
Screen evidence	F2-E03/E04/E09
Simulated study result	F-SIM-02/04/05/06 show status or handoff hesitation and 2-3/5 confidence
Root cause	State, freshness, and destination are separated
Consequence	Delayed action, unsafe assumption, or abandonment

Field	Record
Market gap	Official sports sites rarely combine governed state, freshness, and destination preview in one decision view
Scope	Comparison, preview, alerts, and recovery inside FIFA.com
Out of scope	Transaction, inventory creation, pricing, or partner redesign
PA3 test question	Can users accurately choose buy, wait, register interest, resale, hospitality, or stay?

7.2 TP-CHESS

TP-CHESS: Beginners need one interpretable mistake and one next practice action before advanced analysis.

Field	Record
Affected users	First-time beginners and returning low-analysis learners
Context	After-game review and practice continuation
Screen evidence	C2-E05/E07-E10
Simulated study result	C-SIM-01/02/04/05/06 show wrong paths, vocabulary hesitation, and continuation need
Root cause	Feature/setup organization precedes the learning outcome
Consequence	Expert view without an actionable learning priority
Market gap	Many analysis tools optimize depth before beginner interpretation and continuation
Scope	Beginner preset, one mistake, try move, explanation, practice, optional depth

Field	Record
Out of scope	Engine redesign, coaching guarantees, or removal of expert controls
PA3 test question	Can users explain one mistake and select one relevant practice action?

8. Design rationale and validation target

HCI rationale: Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 - Fundamental Concepts - Usability Dimensions_2.pdf, pp. 62-67, 74-76, 83-84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36; LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

Design inference: The selection balances effectiveness, efficiency, learnability, memorability, errors, satisfaction, status visibility, recognition, progressive disclosure, user control, and error prevention.

Revised Affinity Diagrams

The diagrams separate FIFA and Chess. Each row traces representative raw notes through a small cluster, pattern, problem, severity, and design direction.

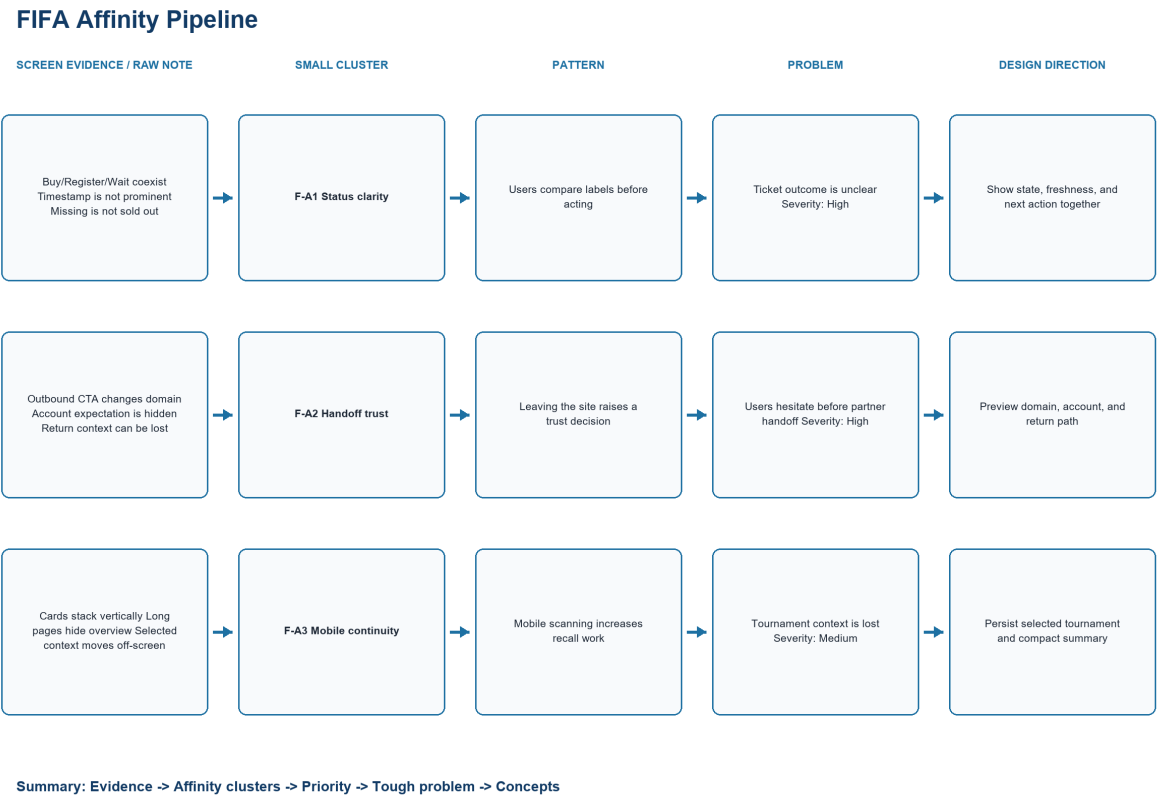


Figure UA-AF-FIFA. FIFA evidence-to-direction affinity diagram
Source: PA2 design QA artifact
Related: Screen evidence -> cluster -> pattern -> problem -> design direction

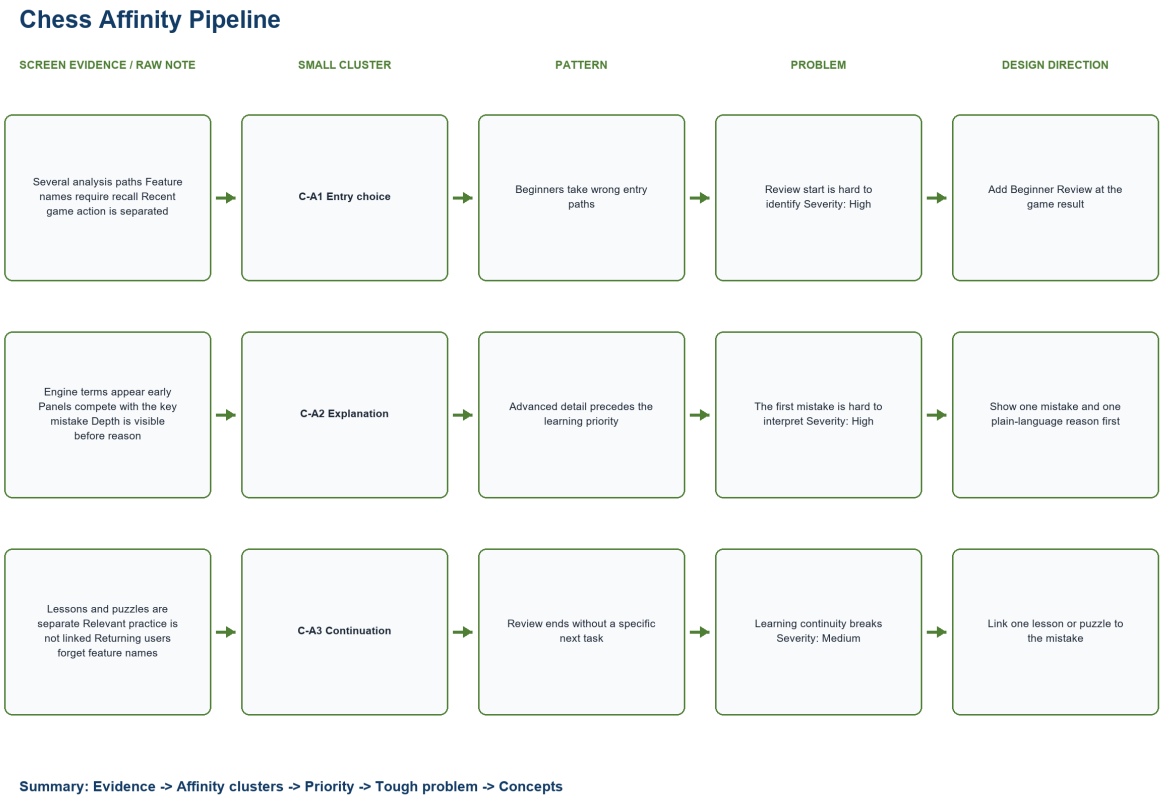


Figure UA-AF-CHESS. Chess evidence-to-direction affinity diagram

Source: PA2 design QA artifact

Related: Screen evidence -> cluster -> pattern -> problem -> design direction

Reading order for both diagrams: SCREEN EVIDENCE / RAW NOTE -> SMALL CLUSTER -> PATTERN -> PROBLEM -> DESIGN DIRECTION. The summary then links evidence to affinity clusters, priority, the tough problem, and concepts.

Practical Difficulties and Lessons Learned

Record 1

Practical difficulty	Seventy-two notes could not remain legible on one page.
How the group handled it	The group split product views and kept only three representative notes in each visible cluster.
Lesson learned	Affinity synthesis should show the transformation, not only the notes.
Impact on the final version	The final figures expose pattern, task, severity, and design implication.

Record 2

Practical difficulty	Use-case and PA1 terms drifted during clustering.
How the group handled it	Stable FIFA and Chess cluster names were used across analysis and proposal sections.
Lesson learned	Terminology is part of traceability.
Impact on the final version	The revised diagrams link each problem to one named design direction.